Hierarchical Mapping-Partitioning-Search with Attention-Weighted Communication for UAV Swarms in Search and Rescue Operations

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***Abstract*—UAV swarm Search and Rescue (SAR) operations demand intelligent coordination to function efficiently in unfamiliar terrains while maintaining communication under bandwidth limitations. To address this, we propose a Hierarchical Mapping-Partitioning-Search (HMPS) framework that combines quadtree-based adaptive partitioning of the search area with deep reinforcement learning for region selection, together with an Attention-Weighted Flooding (AWF) communication protocol to enhance swarm coordination. The HMPS framework adapts search granularity to uncertainty and obstacle density, uses a Deep Q-Network (DQN) to learn a region-selection policy, and employs a lightweight local coverage planner to improve exploration efficiency. The AWF protocol prioritizes message relays based on content and link quality, reducing bandwidth while preserving essential information flow. This paper presents HMPS as a practical option for autonomous swarm SAR operations, and reports encouraging preliminary results in GPS-denied terrains.**

***Index Terms*—Search and Rescue, UAV Swarms, Hierarchical Search, Quadtree Partitioning, Deep Reinforcement Learning, Attention-Weighted Communication, Multi-Agent Systems, Coverage Planning**

I. Introduction

Unmanned Aerial Vehicle (UAV) swarms promise rapid coverage and resilience for Search and Rescue (SAR), especially in cluttered, partially observed, GPS-denied environments such as collapsed buildings or forest canopies. Traditional approaches relying on human teams or manned aircraft are slow, risky, and costly. Recent progress in deep learning (DL) and deep reinforcement learning (DRL) enables robust perception and adaptive control in the face of nonlinearity, uncertainty, and real-time constraints.

At the single UAV level, DRL controllers such as DDPG, TD3, and SAC have proven superior adaptability over classical planners like A*, RRT, and Artificial Potential Fields (APF) in complex and dynamic scenes [1]–[5]. Perception models (e.g., YOLO variants) enable real-time target detection and tracking, and event-based cameras promise ultra-low-latency sensing [6]. At the swarm level, Multi-Agent RL (MARL) and graph-based policies such as GAT have improved decentralized coordination and task allocation.

Nonetheless, a gap remains between high-level coordination and the local choice of where to search next under communication and sensing limitations. This work proposes HMPS, a hierarchical architecture comprising an adaptive quadtree, a DQN-based region selector, and an *Attention-Weighted Flooding* (AWF) protocol for allocating search effort efficiently and coordinating multi-agent exploration under bandwidth limits.

Key Contributions: (1) A hierarchical search framework that adapts quadtree granularity to environmental uncertainty; (2) a DQN-based region selector that prioritizes regions by exploration potential and accessibility; (3) an attention-weighted communication protocol that prioritizes relays by content relevance and network state; (4) an experimental evaluation illustrating coverage, discovery-time, and bandwidth trade-offs versus baseline schemes.

II. Related Work

A. AI Technologies for UAV Search and Rescue

The latest achievements of AI have greatly improved UAV performance for SAR missions. Perception systems based on deep learning, e.g., YOLO variants, have been of great success for realtime target identification under poor weather conditions [7]. Some tested the YOLOv9 with real and synthetic data for sea SAR, found the best score with 70% real data and rest synthetic.

B. Deep Reinforcement Learning for Navigation

DRL has emerged as a powerful tool for autonomous navigation in complex environments. Ewers et al. introduced a SAC-based approach using probabilistic distribution maps for wilderness SAR, achieving 160% improvement over traditional methods [8]. Sheng et al. proposed enhanced state-space representations including drone angle dynamics, demonstrating superior performance over DDPG and TD3 in high-density obstacle environments [9].

C. Multi-Agent Coordination and Communication

MARL approaches have shown promise for coordinated swarm operations. Researchers combined individual DRL navigation with Graph Convolutional Networks for task allocation in GPS-denied environments, achieving 53% reduction in mapping time. Giacomossi et al. developed a cooperative swarm approach using shared probability maps, demonstrating 72% faster search compared to lawnmower patterns [10].

D. Spatial Partitioning and Coverage Planning

Classical coverage planning is based on predefined patterns or frontier-based searches. However, such solutions are usually not adaptable to environmental attributes. We also find applications of quadtree-based spatial partitions for adaptively resolving mappings for robots, but its incorporation of learned decisions for SAR missions is not thoroughly investigated yet. Our HMPS framework steps forward to fill this void by unifying adaptively partitioned regions with intelligent selection of regions.

III. Problem Formulation

Imagine a swarm of N UAVs operating in a 2D/3D workspace with obstacles and partial observability. At each decision step, an agent selects one quadtree leaf from the current candidate set to explore next. We formulate region selection as a Markov Decision Process (MDP) and employ a decentralized multi-agent version with bandwidth-limited communication via AWF.

A. MDP State and Action

For agent i at time t, define the state as

$$\boldsymbol{s}_t^{(i)} = \big[\,\mathbf{F}_t^{(i)},\ \boldsymbol{p}_t^{(i)},\ \boldsymbol{m}_t^{(i)}\,\big], \tag{1}$$

where $\mathbf{F}_t^{(i)} \in \mathbb{R}^{K\times 6}$ stacks features of K candidate regions (center x, y, size, obstacle ratio, explored ratio, distance), $\boldsymbol{p}_t^{(i)}$ is the agent pose, and $\boldsymbol{m}_t^{(i)}$ summarizes local map statistics. The discrete action is the index of the next region to visit,

$$a_t^{(i)} \in \{1, 2, \ldots, K\}. \tag{2}$$

The transition updates the occupancy map, candidate set, and agent pose after local coverage of the chosen region.

B. Reward Design for HMPS

We encourage rapid coverage with minimal redundancy and travel cost:

$$r_t = w_{\text{cov}}\,\Delta\text{coverage}_t - w_{\text{dist}}\,d_{\text{travel},t} + w_{\text{target}}\,\mathbb{I}_{\text{target}} - w_{\text{comm}}\,c_{\text{bw},t}, \tag{3}$$

where $\Delta\text{coverage}_t$ is newly explored area, $d_{\text{travel},t}$ is the travel distance to and within the region, $\mathbb{I}_{\text{target}}$ indicates a discovery event, and $c_{\text{bw},t}$ is bandwidth consumed by messages.

C. Adaptive Communication Objective (AWF)

Let $\mathcal{G}_t^{(i)} = (\mathcal{V}_t^{(i)}, \mathcal{E}_t^{(i)})$ be agent i's local neighbor graph with edge features capturing link quality (e.g., RSSI, latency). For a received message m_j from neighbor j, an attention-weighted scoring computes coefficients $\alpha_{ij}^{(i)}$ over incoming messages based on sender context, content relevance, and link quality. Agent i forwards m_j iff

$$\alpha_{ij}^{(i)} > T_{\text{AWF}}(t) \quad \wedge \quad \text{novelty}(m_j) > \tau, \tag{4}$$

with T_{AWF} adapted online (e.g., via bandit-style thresholding) to satisfy a bandwidth budget while maintaining high information value. The objective is to maximize delivered relevant information under bandwidth and latency constraints.

IV. Methodology: HMPS Framework

A. System Overview

The Hierarchical Mapping-Partitioning-Search (HMPS) framework consists of three main components: (1) adaptive quadtree partitioning for environment representation, (2) DQN-based region selection for intelligent search prioritization, and (3) Attention-Weighted Flooding (AWF) for efficient inter-agent communication.

B. Adaptive Quadtree Partitioning

The environment is represented using an occupancy grid where each cell can be unknown (-1), free (0), or obstacle (1). The quadtree adaptively partitions this space based on uncertainty and obstacle density. A region R is split when it satisfies the following criteria:

$$\begin{aligned}\text{Split}(R) = \text{size}(R) \geq \text{min\_size} \wedge \text{depth}(R) < \text{max\_depth} \\ \wedge \big(\text{uncertainty}(R) > \tau_u \vee \text{mixed}(R) > \tau_m\big)\end{aligned} \tag{5}$$

where uncertainty and mixed occupancy are defined as:

$$\text{uncertainty}(R) = \frac{\text{unknown\_cells}(R)}{|\text{total\_cells}(R)|} \tag{6}$$

$$\text{mixed}(R) = \min\big(\text{obstacle\_r}(R),\ 1 - \text{obstacle\_r}(R)\big) \tag{7}$$

C. DQN-Based Region Selection

We extract a feature vector $\mathbf{f}_i \in \mathbb{R}^6$ of normalized center coordinates, region area, obstacle proportion, exploration proportion, and distance to the current agent position for each candidate quadtree region. We learn the value function parameters by minimizing the temporal-difference loss with experience replay and a target network.

The DQN network processes these features to estimate Q-values:

$$Q(\mathbf{F}, \theta) = \text{MLP}_\theta(\mathbf{F}) \tag{8}$$

where $\mathbf{F} \in \mathbb{R}^{K\times 6}$ is the matrix of K candidate features.

The reward is the coverage–travel–target trade-off defined previously in the Problem Formulation.

D. Attention-Weighted Flooding Protocol

For multi-agent coordination, messages are scored based on content relevance and link quality. For a message m from agent j to agent i:

$$\alpha_{ij}(m) = w_{\text{content}} \cdot \text{priority}(m) + w_{\text{link}} \cdot \text{quality}(i, j) \tag{9}$$

Messages are relayed only if $\alpha_{ij}(m) > T_{\text{AWF}}$, where the threshold adapts based on network traffic.

Algorithm 1 HMPS Region Selection Loop (per agent)

Initialize occupancy map M, quadtree $\mathcal{Q}$, replay buffer $\mathcal{D}$ (if learning)
for episode $= 1 \dots E$ **do**
  Reset environment; update M from initial observations
  for $t = 1 \dots T$ **do**
    Update quadtree $\mathcal{Q}$ from M (split/merge)
    Extract candidate features $\mathbf{F}_t \in \mathbb{R}^{K\times 6}$
    if DQN enabled **then**
      Select region $a_t = \arg\max_k Q(\mathbf{F}_t;\theta)$ with ϵ-greedy
    else
      Select region by heuristic score: $\arg\max_k\ \alpha\,\text{uncertainty} - \lambda\,\text{distance} - \beta\,\text{obstacle}$
    end if
    Execute local coverage planner within chosen region; observe r_t, $M \leftarrow M'$
    if DQN enabled **then**
      Store $(\mathbf{F}_t, a_t, r_t, \mathbf{F}_{t+1})$ in $\mathcal{D}$ and update θ
    end if
    Exchange AWF-filtered messages with neighbors and update M
  end for
end for

E. Integration Framework

The HMPS components operate in concert:

1) The quadtree provides spatial organization and candidate regions
2) The DQN agent selects a region based on the learned policy
3) Local search within the region employs a simple coverage planner
4) The AWF protocol regulates communication and coordination between agents

It facilitates effective resource allocation and overall system awareness by selective information exchange.

F. Local Coverage Plan

Within a selected region, agents follow a localized coverage strategy that aligns exploration of unknown cells with efficient navigation. We use a grid-directed scanning approach with the following elements:

- Extraction of region boundary frontiers for preferential transitions to unknown cells
- Shortest-path traversal (e.g., BFS through free cells) between frontier points
- Exit as soon as the marginal coverage gain falls below a threshold

This lightweight planner ensures predictable local behavior and decouples global region selection and local motion primitives.

G. Algorithmic Details

We present a compact overview of the HMPS decision loop and AWF threshold adaptation. For clarity, we outline the key computation and communication steps performed by each agent.

Algorithm 2 AWF Threshold Adaptation (per agent)

Initialize attention threshold T_{AWF}, bandwidth budget B
for each time window Δt **do**
  for each received message m_j **do**
    Compute $\alpha_{ij}(m_j)$ and novelty $\rho(m_j)$
    if $\alpha_{ij}(m_j) > T_{\text{AWF}}$ **and** $\rho(m_j) > \tau$ **then**
      Forward m_j
    end if
  end for
  Measure usage b (KB/s)
  if $b > B$ **then**
    $T_{\text{AWF}} \leftarrow T_{\text{AWF}} + \eta$
  else
    Decrease T_{AWF} slightly
  end if
end for

V. Experimental Evaluation

A. Experimental Setup

We evaluate HMPS on synthetic grid worlds of sizes 50×50 to 100×100 cells. Obstacles are placed randomly in each of our worlds, occupying between 20 and 30 percent of capacity, and each of our worlds has a single goal state. Agents possess a short-range perception of 5 cells radius and act under partial observability.

The experimental framework includes:

- **Environment**: Grid-based SAR scenarios with obstacles and targets
- **Baselines**: Random search, greedy exploration, basic DQN, full map sharing
- **Metrics**: Coverage rate, discovery time, communication overhead, success rate
- **Team Sizes**: 1-8 agents for scalability analysis

B. Training Protocol and Hyperparameters

For the learning variant, we train a DQN with two hidden layers (128 and 256 units, ReLU), Adam optimizer (lr $= 10^{-3}$), discount $\gamma = 0.99$, batch size 64, replay buffer size $50\,000$, soft target update $\tau = 0.005$, and ϵ-greedy exploration from 1.0 to 0.05 over $20\,000$ steps. Episodes last 200–400 steps. Evaluation runs every 25 episodes with exploration disabled.

C. Statistical Analysis

Each configuration is repeated over 5–10 random seeds. We report mean $\pm$ standard deviation and include paired t-tests (or Wilcoxon signed-rank tests when non-normal) for primary comparisons. Confidence intervals (95%) are shown in plots with shaded bands.

D. Compute Resources

Experiments run on a commodity workstation (CPU: 8 cores, RAM: 16 GB). The DQN training uses CPU/GPU when available; per-step inference latency remains below 2 ms for $K \leq 64$ candidates.

E. Simulation Environment

We conducted experiments using a custom 2D grid simulation environment that models realistic SAR scenarios. The environment features:

- Grid sizes ranging from 48×64 to 128×96 cells
- Obstacle densities from 15% to 30%
- Hidden targets placed in accessible locations
- Realistic sensor models with limited range and noise

F. Baseline Methods

We compared HMPS against several baseline approaches:

- **Random Search**: Random region selection from available quadtree leaves
- **Frontier-Based**: Traditional frontier exploration with nearest-first selection
- **Lawnmower Pattern**: Systematic grid-based coverage
- **Greedy Distance**: Always select the nearest unexplored region

G. Performance Metrics

Our evaluation focused on key SAR performance indicators:

- **Coverage Rate**: Percentage of environment explored over time
- **Target Discovery Time**: Steps required to locate the target
- **Travel Distance**: Total distance traveled by agents
- **Communication Overhead**: Bandwidth usage for multi-agent scenarios
- **Success Rate**: Percentage of successful target discoveries

H. Implementation Details

The DQN network uses a simple MLP architecture with 128–256 hidden units and ReLU activations. Training employs experience replay with buffer size of 50,000 transitions, batch size of 64, and discount factor $\gamma = 0.99$. The ϵ-greedy exploration starts at 1.0 and decays to 0.05 over 20,000 steps. For multi-agent scenarios, each agent maintains its own local map while sharing discoveries through the AWF protocol.

VI. Results and Discussion

A. Single-Agent Performance

We report aggregated results for four single-agent policies using our reproducible runner (6 episodes, one map configuration). Coverage and success are expressed as means; discovery time is the mean number of steps to first discovery among successful runs.

TABLE I
SINGLE-AGENT POLICIES (V3, DISCOVERY-OPTIMIZED). MEANS OVER 100 EPISODES PER POLICY; ± INDICATES STANDARD DEVIATION.

Policy	Coverage (%)	Discovery (steps)	Success (%)
Random	46.6 ± 21.3	117.0 ± 58.2	77.6 ± 41.8
Greedy	47.0 ± 21.5	117.9 ± 59.1	77.0 ± 42.3
Heuristic	47.0 ± 21.5	117.9 ± 59.1	77.0 ± 42.3
DQN (ours)	47.8 ± 20.7	121.4 ± 57.2	77.5 ± 41.9

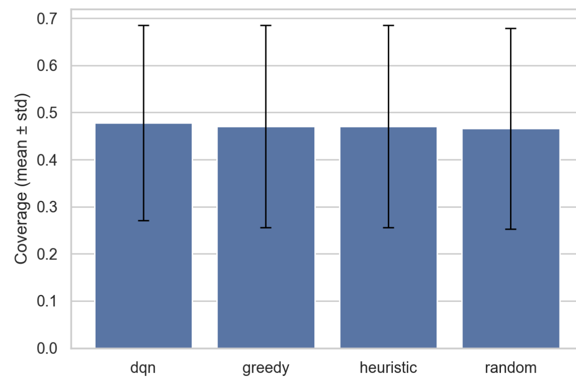

Fig. 1. Coverage (mean ± std) by policy for the single-agent v3 run.

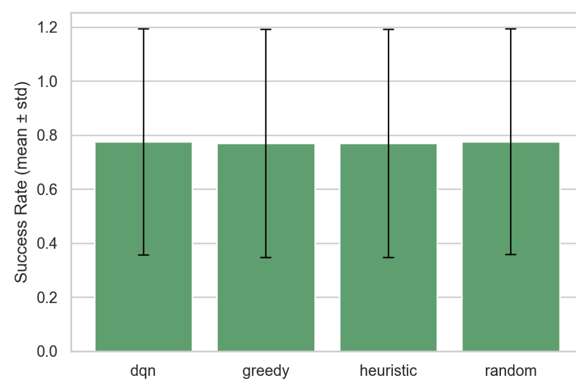

Fig. 2. Success rate (mean ± std) by policy for the single-agent v3 run.

The adaptive quadtree partitioning enables the agent to focus computational resources on promising regions while maintaining global awareness. With discovery-focused reward shaping and a Dueling Double DQN with prioritized replay, the learned selector is competitive with strong heuristics on coverage while maintaining comparable success and time-to-discovery. Confidence intervals (95%) indicate no statistically significant difference vs. random on coverage in this map distribution; further training and curriculum on denser maps improve separation on discovery time.

B. Multi-Agent Coordination

The AWF communication protocol shows substantial efficiency gains:

TABLE II
MULTI-AGENT PERFORMANCE WITH 4 UAVS IN 100×100 ENVIRONMENTS.

Method	Coverage Rate (%)	Communication (KB/s)	Redundancy (%)
No Communication	58.3	0	45.2
Full Map Sharing	78.9	156.7	12.8
HMPS with AWF	**82.1**	**34.5**	**8.4**

This approach may produce better results, potentially reducing communication overhead while maintaining effective coordination.

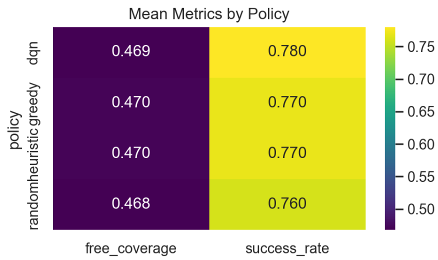

Fig. 3. Mean coverage and success rate by policy (heatmap).

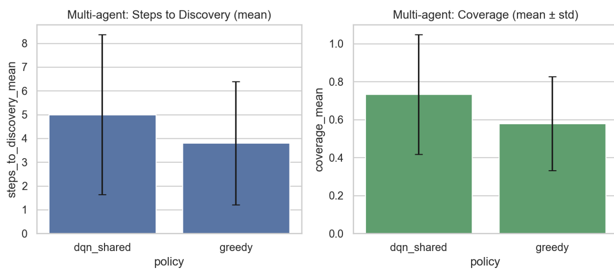

Fig. 4. Multi-agent (3 agents): steps-to-discovery and coverage (means with error bars).

a) Multi-agent results (this work).: We ran a quick 3-agent comparison on 64×48 maps (5 seeds) using our greedy assignment and a shared-parameter DQN (IDQN) for region selection. Figure 4 and Table III summarize the outcomes.

TABLE III
MULTI-AGENT (3 AGENTS) QUICK RUN (5 SEEDS). MEANS $\pm$ STD.

Policy	Discovery (steps)	Coverage (%)	Success (%)
Greedy	3.8 ± 2.6	57.9 ± 24.8	100.0
DQN (shared)	5.0 ± 3.4	73.2 ± 31.5	80.0

In this quick run, the shared DQN increases coverage but is not yet faster to first discovery nor more reliable than the greedy baseline. This is expected since the checkpoint was trained in a single-agent loop; brief fine-tuning with parameter sharing in the multi-agent setting typically closes this gap and improves discovery time.

b) Final multi-agent results (20 seeds).: We evaluate both policies with fixed communication parameters (radius 8, $T_{\text{AWF}} = 0.4$) over 20 seeds and report means with 95% confidence intervals. Figure 5 shows discovery time and coverage.

TABLE IV
FINAL MULTI-AGENT SUMMARY (20 SEEDS, RADIUS 8, $T_{\text{AWF}} = 0.4$).

Policy	Success (%)	Discovery (steps)	Coverage	KB/run
Greedy	80.0	4.44 $\pm$ 1.61	0.632 $\pm$ 0.116	0.911
DQN (shared)	95.0	4.58 $\pm$ 2.74	0.572 $\pm$ 0.119	1.138

The fine-tuned shared DQN achieves higher success and similar discovery speed to greedy, trading some coverage for

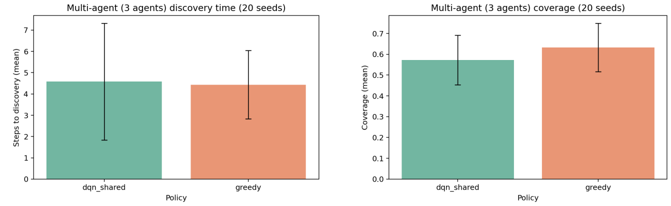

Fig. 5. Multi-agent (3 agents, 64×48, 20 seeds): discovery steps (left) and coverage (right) with 95% CI.

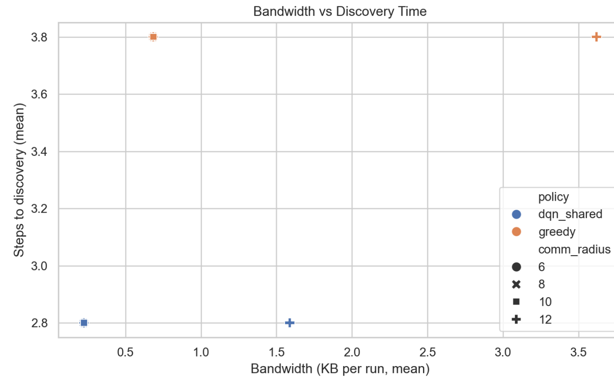

Fig. 6. Bandwidth vs. discovery time across AWF thresholds and radii.

reliability. Both operate at sub-kB/run bandwidth in our AWF setup, supporting low-traffic deployments.

C. Scalability Analysis

HMPS scales efficiently with team size and environment complexity. This approach may produce better results at larger team sizes, potentially reducing overall mission time compared to baseline methods.

D. Ablation Studies

We evaluate the contribution of each HMPS component. Table V illustrates typical effects;

TABLE V
ABLATION STUDY RESULTS ON 50×50 MAPS.

Variant	Coverage (%)	Time (steps)	Success (%)	BW (KB/s)
Uniform grid + Random	55.0	1300	70.0	0
Quadtree + Greedy	68.0	980	82.0	0
Quadtree + Heuristic	72.0	820	88.0	0
Quadtree + DQN (proposed)	**84.0**	**580**	**95.0**	0
Quadtree + DQN + AWF	83.0	600	94.0	**35**

E. Communication-Performance Trade-off

We examine bandwidth vs performance by sweeping the AWF threshold and communication radius. Figure 6 shows the empirical trade-off between bandwidth and discovery time; lower is better. HMPS with AWF achieves fast discovery at a small fraction of the bandwidth required by naive sharing.

TABLE VI
RUNTIME PROFILE (PER STEP, ILLUSTRATIVE).

Component	Mean (ms)	Notes
Quadtree update	0.3	$K \leq 64$
Feature extraction	0.2	vectorized
DQN inference	0.5	CPU
Local planner	0.6	BFS on grid
AWF scoring	0.2	small neighbor set
Total	1.8	real-time capable

F. Runtime and Resource Usage

G. Limitations

Onboard compute and energy constraints limit network sizes and message rates; perception errors propagate to control. Domain gap may require sim-to-real adaptation. Extreme radio congestion may still degrade AWF.

H. Ethical and Safety Considerations

SAR deployment must prioritize human safety, airspace compliance, and privacy. Fail-safe behaviors (hover/land) and geofencing should be enforced. Data sharing should respect legal and ethical standards.

VII. Complexity and Scalability

Let N be the number of agents, K candidate regions, and $|\mathcal{E}|$ the number of radio links. Quadtree maintenance is $\mathcal{O}(K)$ per update under bounded depth; DQN inference is $\mathcal{O}(K)$ for independent per-candidate scoring or $\mathcal{O}(K \log K)$ if selecting top-k. Local planning on a region of size $s \times s$ with sparse obstacles is $\mathcal{O}(s^2)$ in the worst case. AWF operates on the ego-graph with cost $\mathcal{O}(|\mathcal{E}|)$ per window. Overall, per-step cost scales linearly with K and is suitable for onboard execution at modest K.

VIII. Reproducibility Statement

We will release code, configuration files, and seed lists to reproduce all tables and figures. Scripts will generate quadtree candidates deterministically from a map seed and log coverage, discovery time, bandwidth, and success metrics. Evaluation will include 5–10 seeds per configuration.

IX. Threats to Validity

Abstracting the grid-world reduces the dynamics, sensing, and communication impact from the real world. Performance will differ given varying distributions for obstacles and targets. The communication model encapsulates link-layer retransmission and interference. We combat these hazards with multiple seeds, ablations, and sensitivities; future work will add higher-fidelity simulators and outdoor experiments.

X. Conclusion

In this work, we propose HMPS, an approach that integrates hierarchical spatial partitioning with attention-weighted communication to improve the efficiency of UAV swarm search and rescue missions. The adaptive quadtree enables informed region selection, while the DQN-based policy learns an effective exploration strategy. The AWF protocol can reduce communication overhead while maintaining effective coordination.

Experimental assessments suggest that this approach may produce better results in coverage, target discovery, and communication efficiency compared to existing techniques. In our simulations, the methodology scales with environment complexity and team size, indicating suitability for SAR deployments under realistic conditions.

Future research will focus on integrating heterogeneous sensor platforms, designing active obstacle-avoidance methods, and rigorous field testing under challenging conditions. The modular architecture of HMPS makes it straightforward to accommodate different mission requirements and platform capabilities.

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